

In-Class Worksheet

Probability and Conditional Probability

Name: _____

Date: _____

Course: H524 Introduction to Biostatistics

Topic: Probability, Conditional Probability, and Diagnostic Testing

Instructions

Work with a partner to solve the following problems. Use the 2x2 table approach with natural frequencies (concrete counts) rather than formulas whenever possible. Show all your work.

1 Problem 1: Mammogram Screening (Based on Gigerenzer's Study)

A 45-year-old woman with no symptoms or family history of breast cancer has a positive mammogram. What is the probability that she actually has breast cancer?

Given information:

- The prevalence of breast cancer in this age group is 1.2%
- If a woman has breast cancer, the probability is 85% that she will have a positive mammogram (sensitivity)
- If a woman does not have breast cancer, the probability is 9% that she will still have a positive mammogram (false positive rate)

Part A: Set up a 2x2 table assuming 10,000 women are screened. Fill in all cells:

	Cancer +	Cancer -	Total
Test +			
Test -			
Total			10,000

Work space for calculations:

Part B: Calculate the Positive Predictive Value (PPV): $\Pr(\text{Cancer} \mid \text{Test}+)$

Part C: Interpret your result. Why is the PPV so much lower than the sensitivity?

2 Problem 2: Strep Throat Testing

A rapid strep test has 85% sensitivity and 95% specificity. Among patients presenting with sore throat, 15% actually have strep throat.

Part A: Create a 2x2 table for 1,000 patients with sore throat:

	Strep +	Strep -	Total
Test +			
Test -			
Total			1,000

Part B: Calculate the Negative Predictive Value (NPV): $\Pr(\text{No Strep} \mid \text{Test-})$

Part C: What is the probability that a patient with a negative test actually has strep (false negative)?

3 Problem 3: Smoking and Heart Disease

In a study of 1,000 people, there were 200 smokers and 800 non-smokers. During the 10-year follow-up, 30 smokers and 20 non-smokers developed heart disease.

Part A: Create a 2x2 table:

	Heart Disease	No Disease	Total
Smokers			
Non-smokers			
Total			1,000

Part B: Calculate the risk of heart disease in smokers:

Part C: Calculate the risk of heart disease in non-smokers:

Part D: Calculate the Relative Risk (RR):

Part E: Interpret the relative risk in plain language:

4 Problem 4: Binomial Probability

A new treatment has a 60% success rate.

Part A: If 10 patients receive the treatment, what is the probability that at least 8 of them respond successfully? Use R or a binomial probability table.

Part B: What is the expected number of successful responses out of 10 patients?

5 Problem 5: Normal Distribution

Cholesterol levels in adults are normally distributed with mean 200 mg/dL and standard deviation 40 mg/dL.

Part A: What proportion of adults have cholesterol levels between 180 and 220 mg/dL?

Part B: What cholesterol level corresponds to the 90th percentile?